PHOT 451: Microscale optical system design
LECTURE 04

Michaël Barbier, Fall semester (2026-2027)

OVERVIEW OF THE COURSE

week	Topic
Week 1	Introduction to micro-scale optical components
Week 2	Light propagation in free space
Week 3	Geometric optics and raycasting
Week 4	Diffraction limit & Abberations
Week 5	Beam propagation
Week 6	Refractive optical elements Microlenses
Week 7	Blazed Fresnel lenses
Week 8	Digital lenses
Week 9	Diffractive optical elements
Week 10	**Project presentations**
Week 11	Wave guides and beam propagation
Week 12	Wave mixing
Week 13	photonic crystals, Gratings, periodic structures
Week 14	Whole optical system optimization

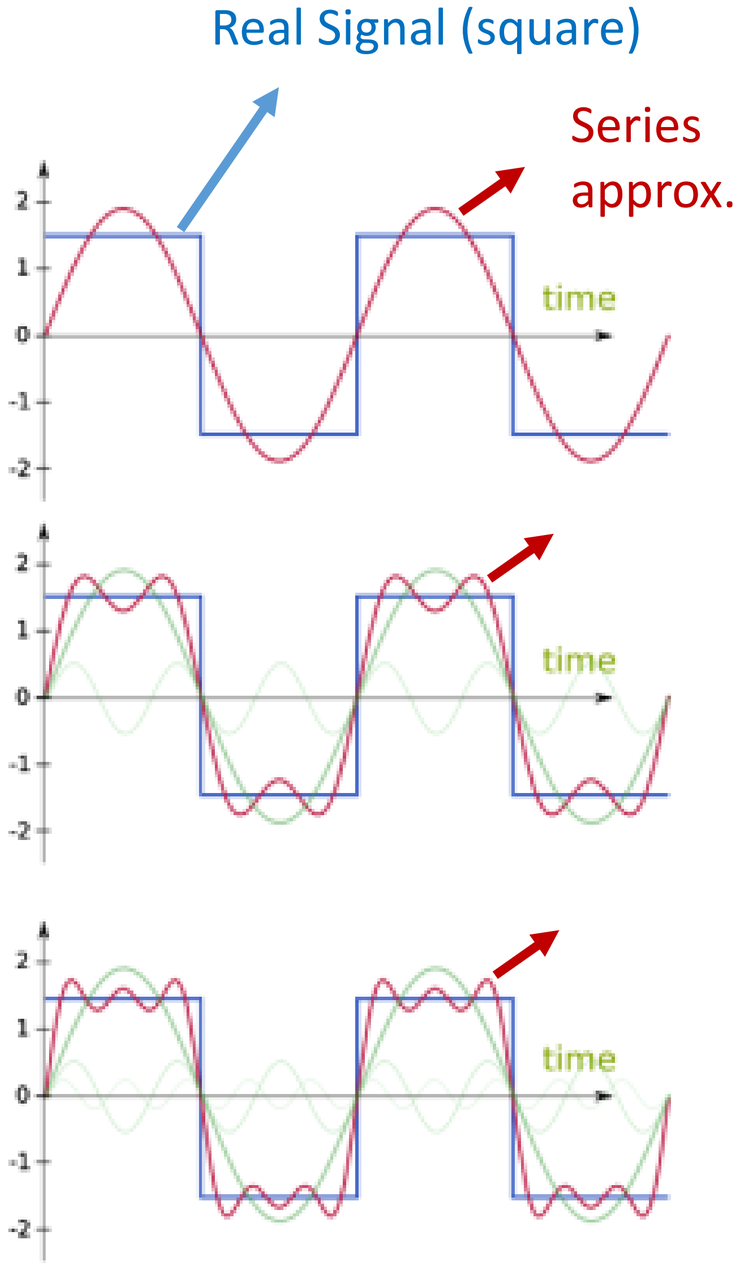
The Diffraction Limit

THE FOURIER SERIES

- Express signals in cosine + sines

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi n}{L} x\right) + b_n \sin\left(\frac{2\pi n}{L} x\right) \right]$$

- Examples
  - Square wave
  - Sawtooth



Adapted from: Lê Nguyên Hoang, Fourier Analysis: Signals and Frequencies, Science4All

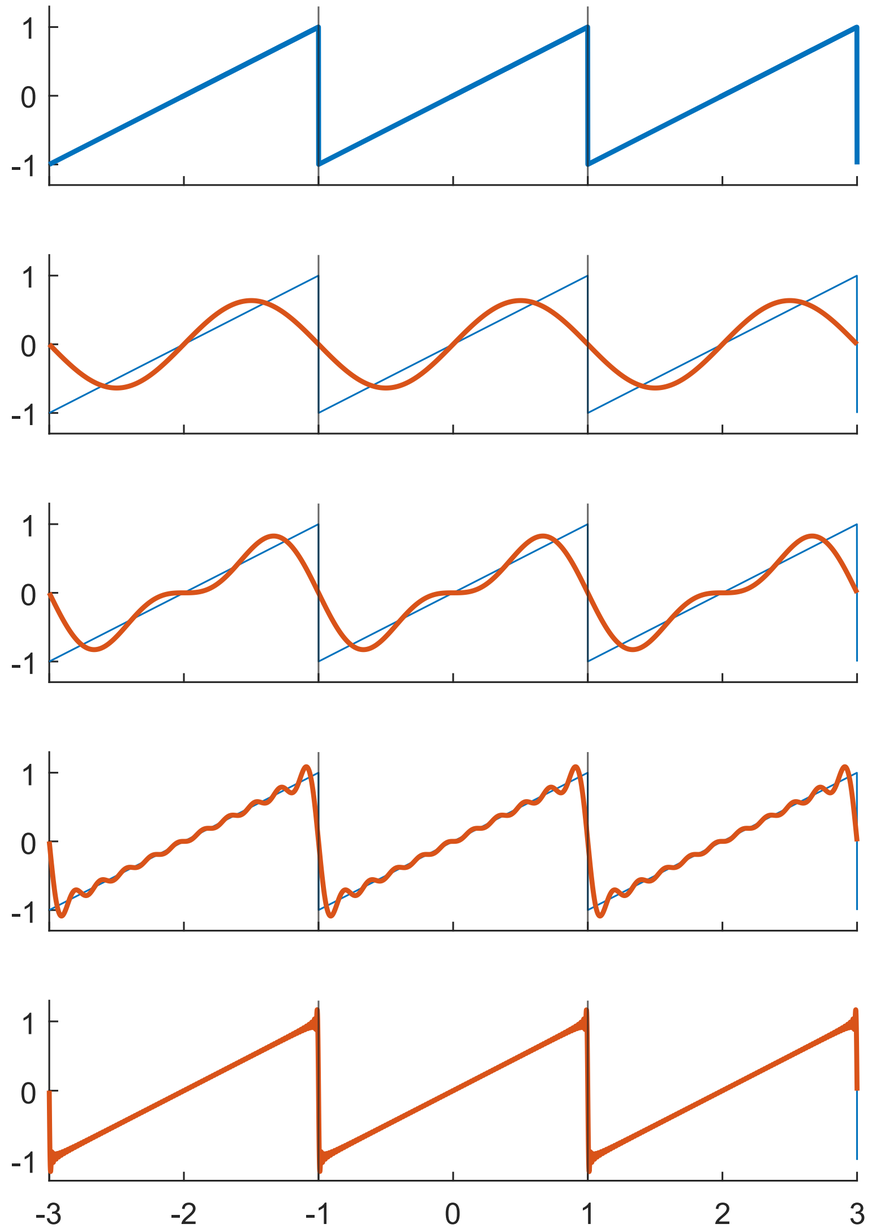
THE FOURIER SERIES

- Express signals in cosine + sines

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi n}{L} x\right) + b_n \sin\left(\frac{2\pi n}{L} x\right) \right]$$

- Examples
  - Square wave
  - Sawtooth

- Artifacts
  - Larger when series finite
  - Can persist for infinite series



THE FOURIER SERIES: EXPONENTIALS

- Fourier series:

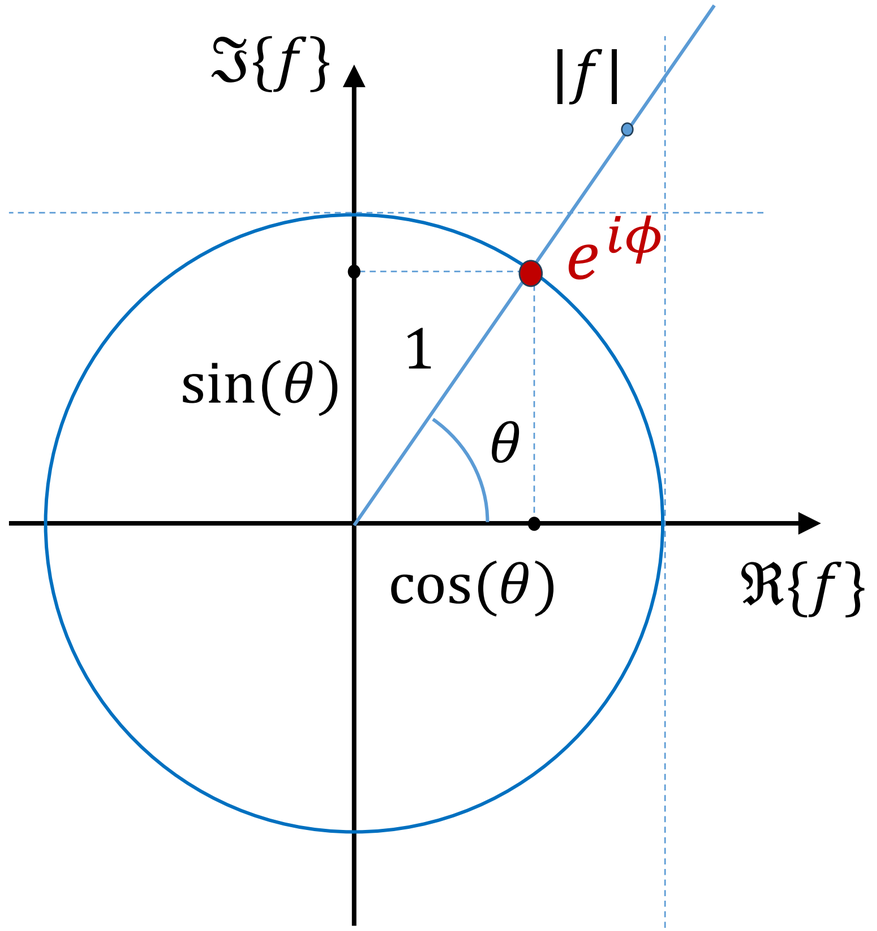
$$f(x) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{2\pi n}{L}x\right) + b_n \sin\left(\frac{2\pi n}{L}x\right)\right]$$

Alternative complex series using:

$$e^{i\phi} = \cos(\phi) + i\,\sin(\phi)$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n \exp\left(\frac{2\pi n}{L}x\right)$$

$$f(x) = a + i\,b = \Re\{f\} + i\,\Im\{f\}$$



FOURIER SERIES: TOWARDS FOURIER TRANSFORM

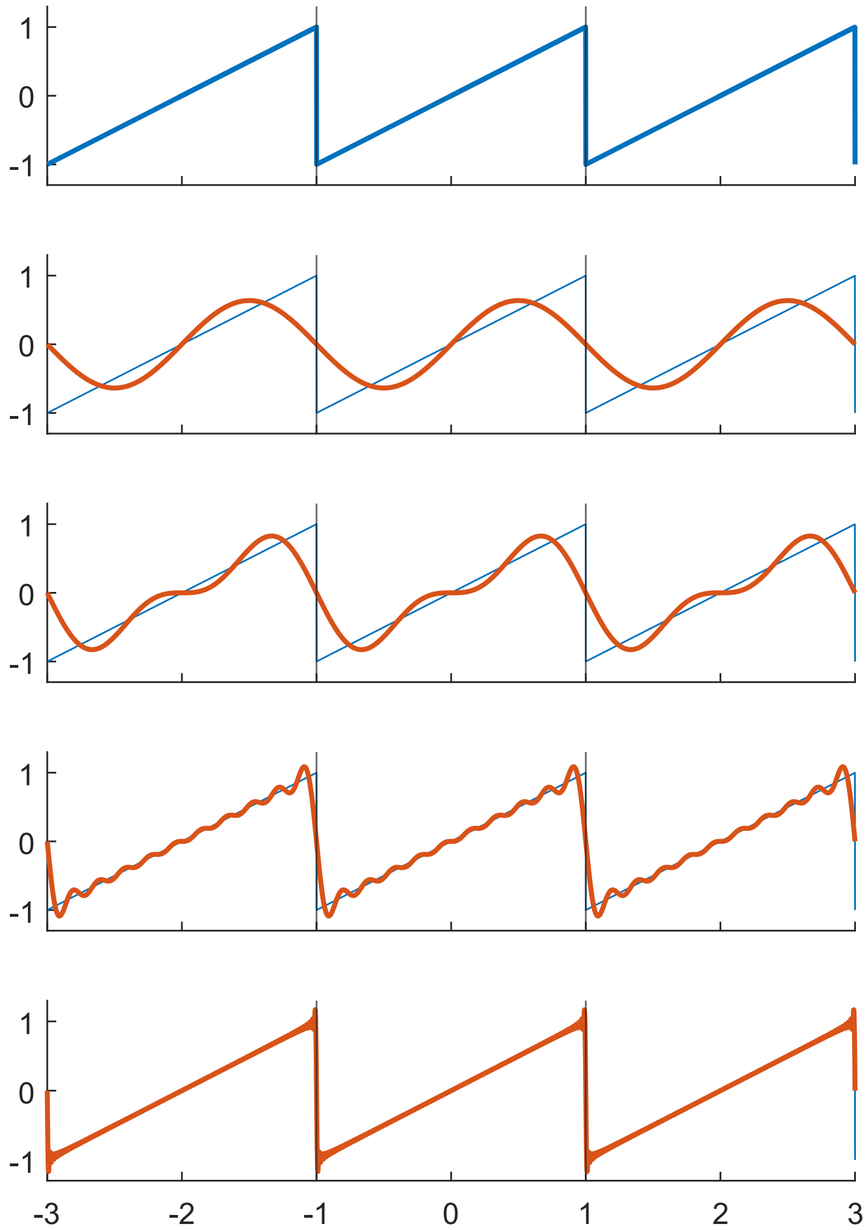
- Series can only represent periodic waves?

$$f(x) = \sum_{n=-\infty}^{\infty} c_n \exp\left(\frac{2\pi n}{L} x\right) = \sum_{n=-\infty}^{\infty} c_n \exp(k_n x)$$

- Problem: finite period L

- Increase $L \to \infty$

$$\mathrm{k_n} = \frac{2\pi n}{L} \Rightarrow \quad dk = \frac{2\pi}{L}$$



ALL SIGNAL SHAPES: THE FOURIER TRANSFORM

Fourier series

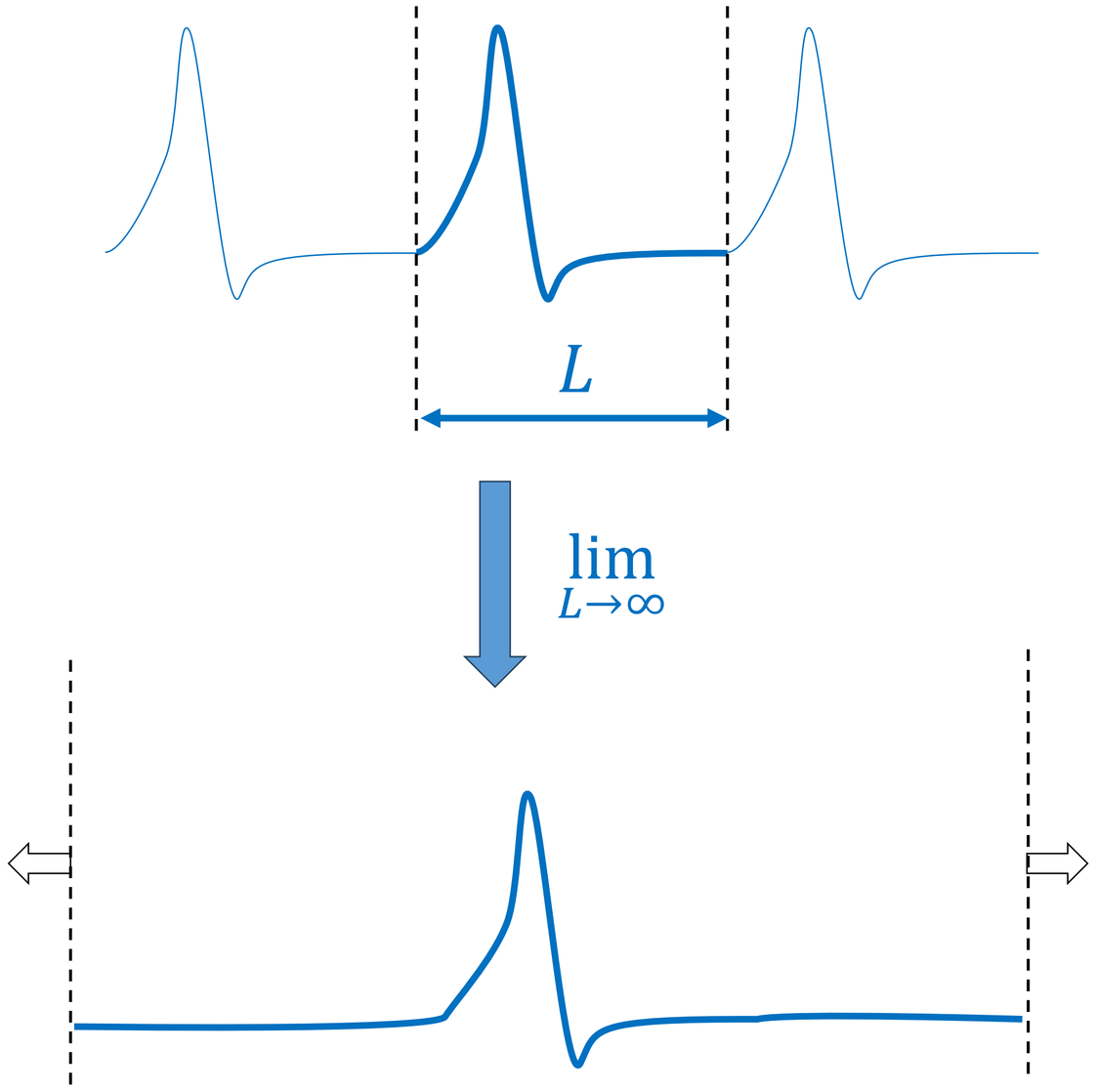
$$f(x) = \sum_{n=-\infty}^{\infty} c_n \exp(k_n x)$$

$$\lim_{L\to\infty} \mathrm{k_n} = \frac{2\pi n}{L} \quad \rightarrow \quad dk = \frac{2\pi}{L}$$

Fourier integral

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx}\, dk$$

$$k_n \rightarrow k, \qquad c_n \rightarrow \tilde{f}(k)$$



THE FOURIER TRANSFORM

- Complex form

$$e^{i\phi} = \cos(\phi) + i\,\sin(\phi)$$

- Forward Spatial Fourier transform

$$f(x) = F_k^{-1}[\tilde{f}(k)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx}\, dk$$

- Inverse Spatial Fourier transform

$$\tilde{f}(k) = F_x[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-ikx}\, dx$$

THE FOURIER TRANSFORM

- Spatial Fourier transform

$$f(x) = F_k^{-1}[\tilde{f}(k)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx}\, dk$$

$$\tilde{f}(k) = F_x[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-ikx}\, dx$$

- Frequency Fourier transform

$$f(t) = F_\omega^{-1}[\tilde{f}(\omega)] = \int_{-\infty}^{\infty} \tilde{f}(\omega) e^{-i\omega t}\, d\omega$$

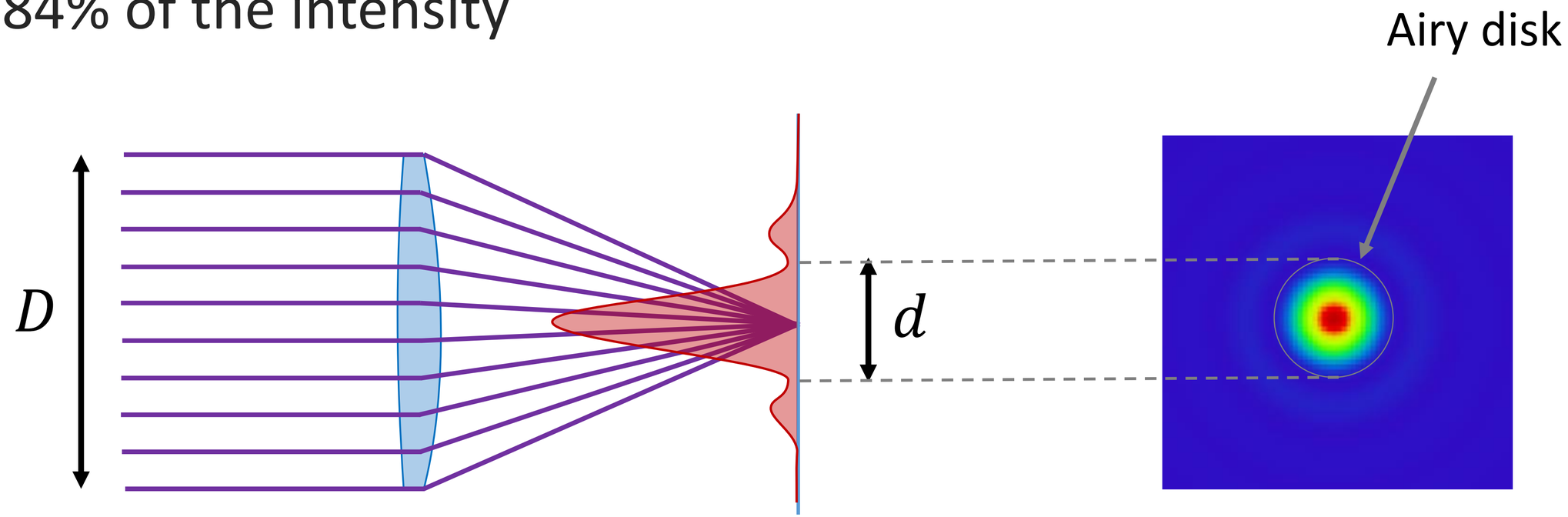
$$\tilde{f}(\omega) = F_t[f(t)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t)\, e^{i\omega t}\, dt$$

THE POINT SPREAD FUNCTION

- Airy disk diameter

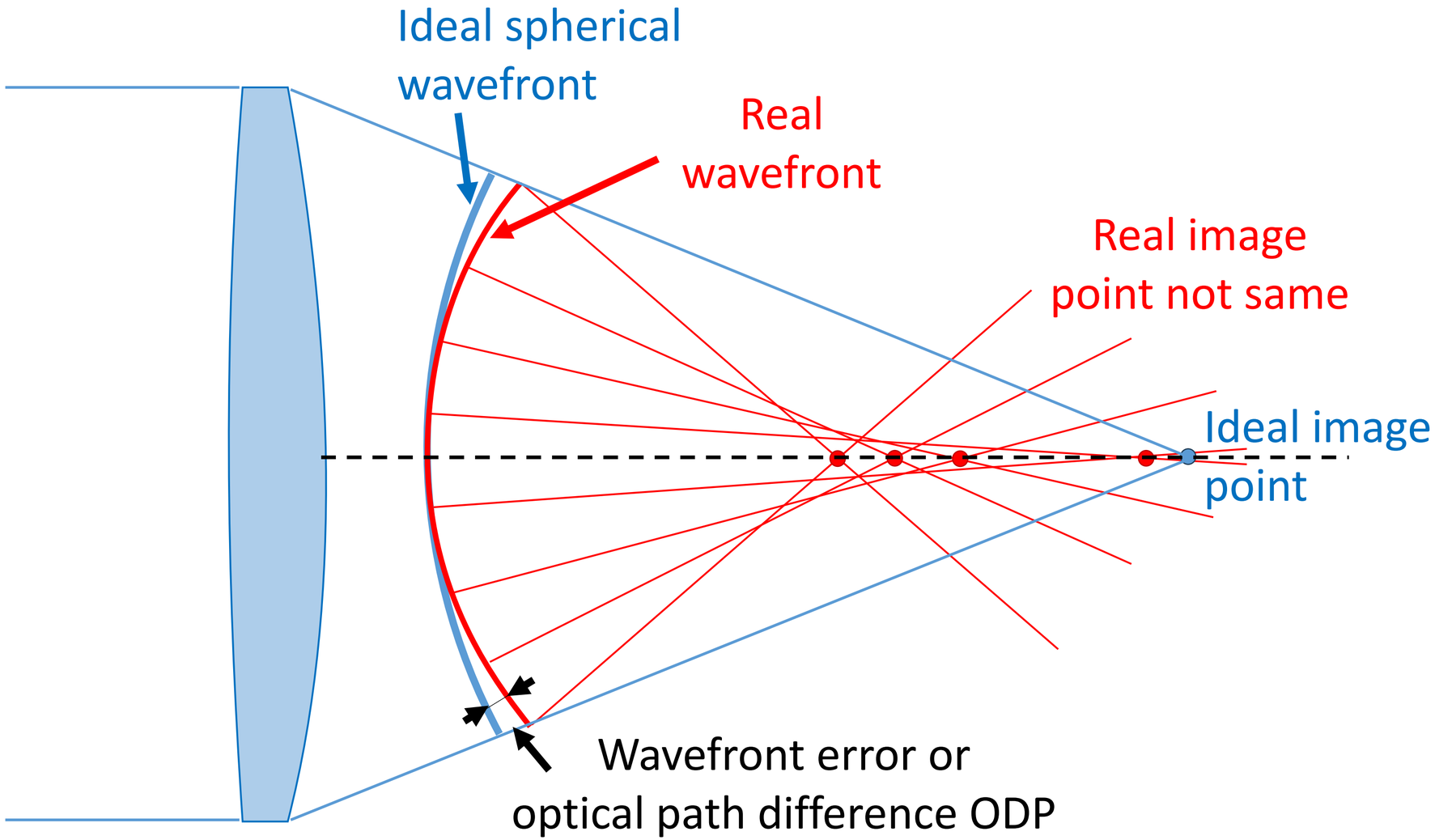
$$d = 2.44\,\lambda\frac{f}{D}$$

- First ring of the Point Spread Function 84% of the intensity

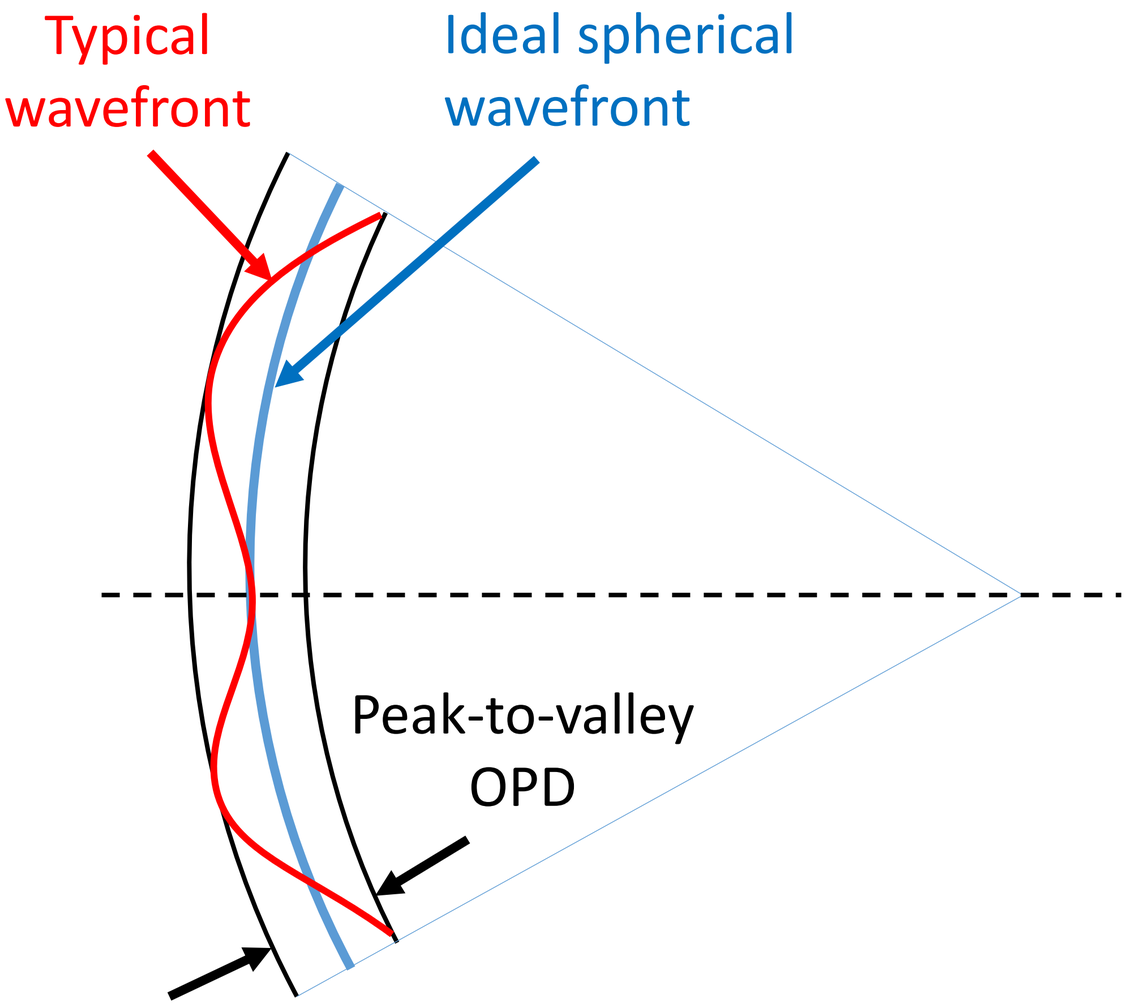


Aberrations

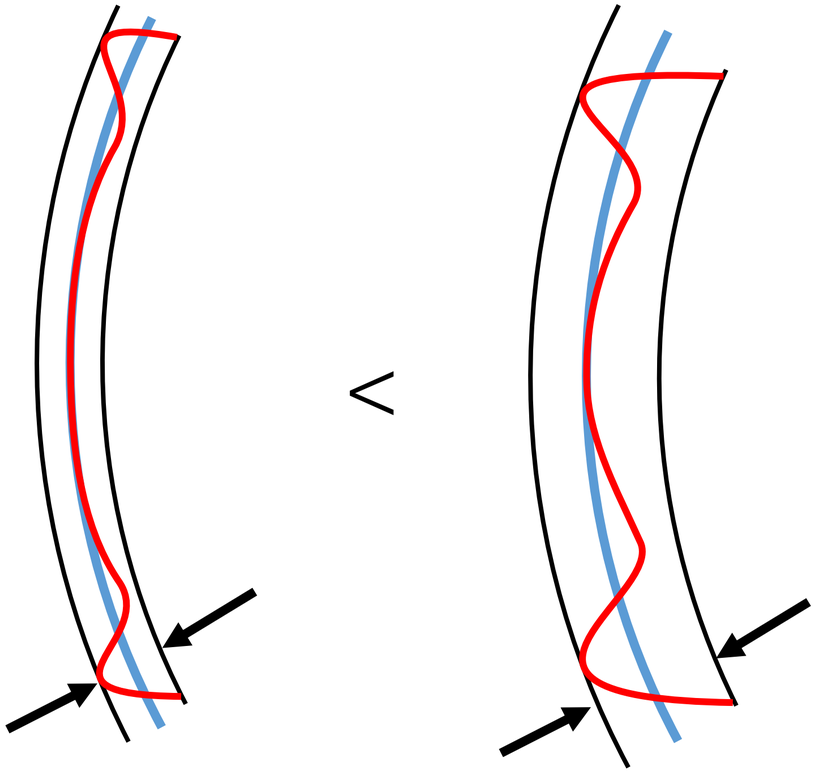
OPTICAL PATH DIFFERENCE



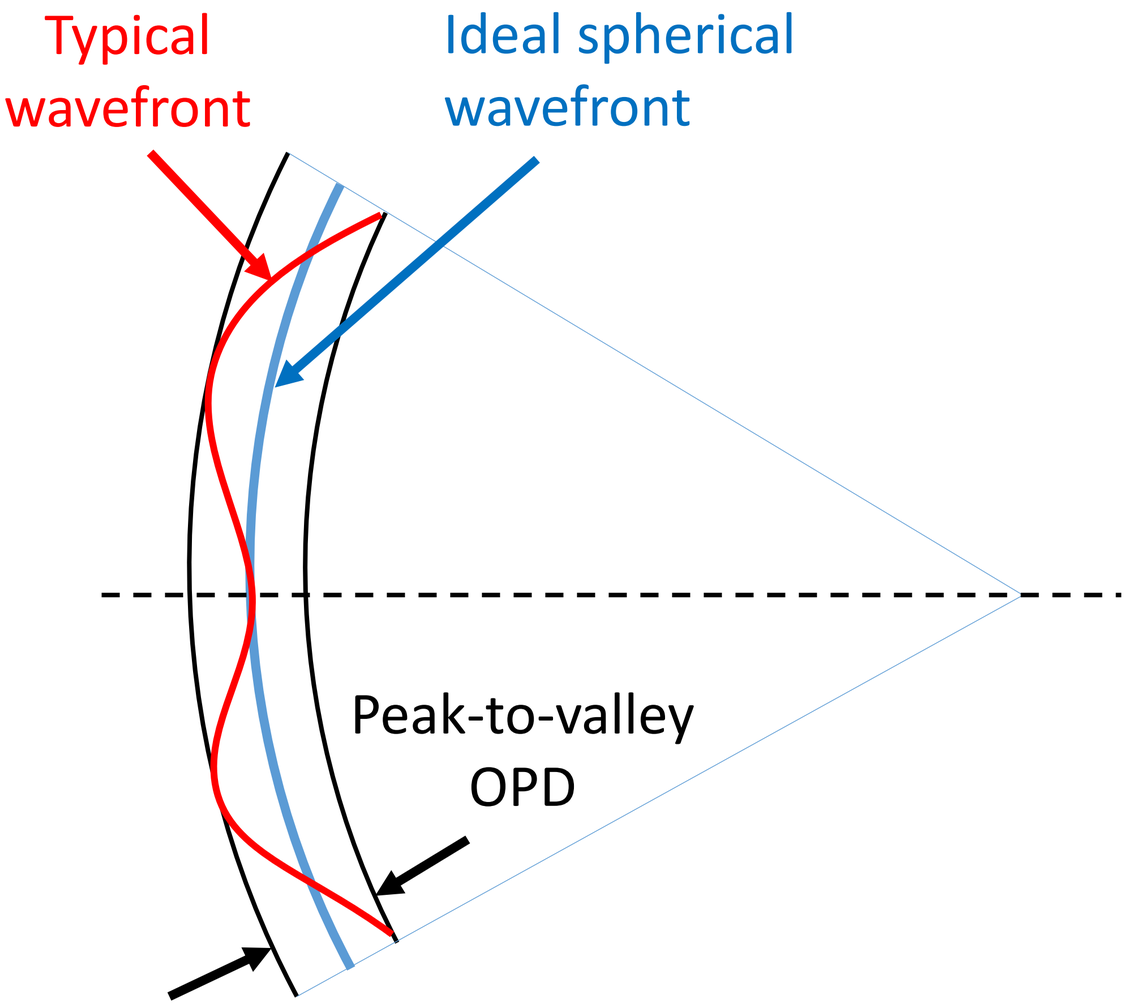
OPTICAL PATH DIFFERENCE



Peak-to-valley OPD

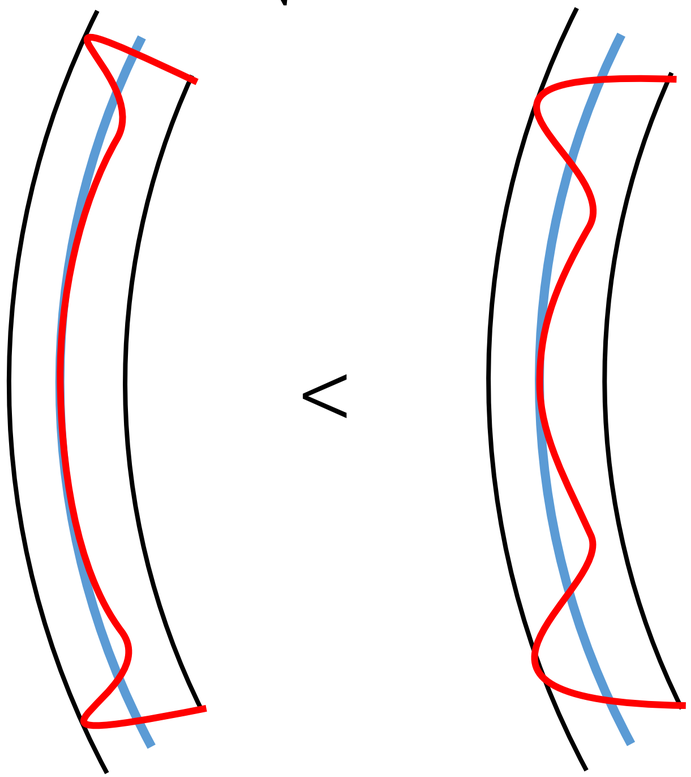


OPTICAL PATH DIFFERENCE

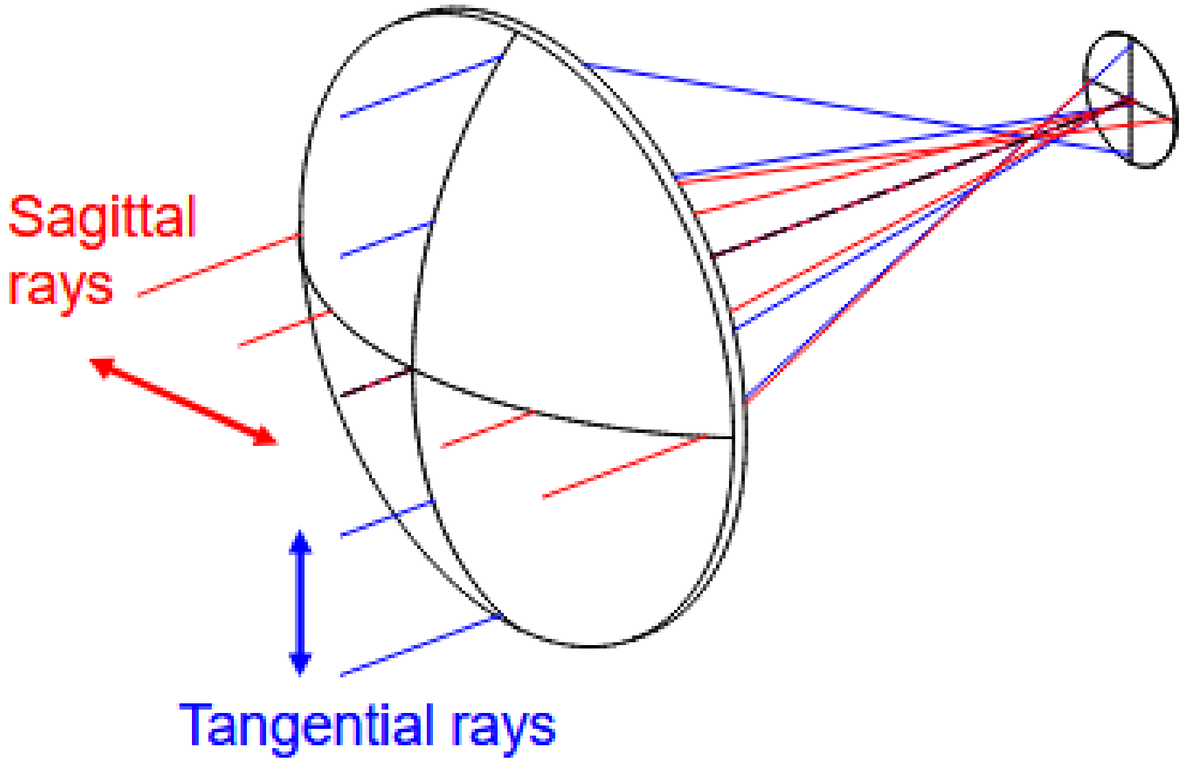


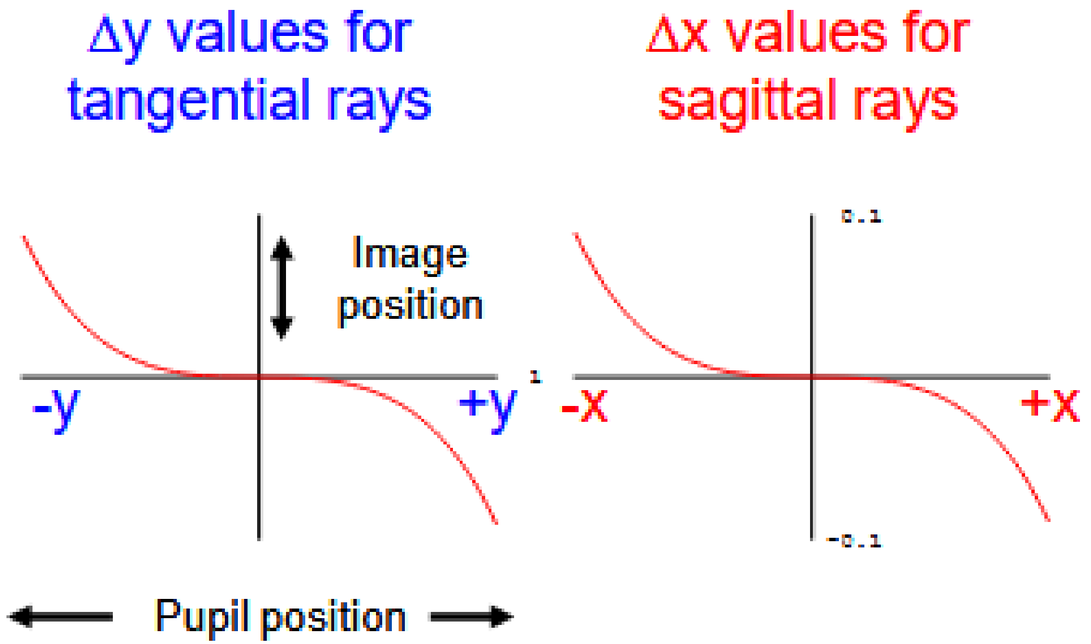
Root Mean Square
RMS OPD

$$\text{RMS} = \sqrt{\sum_j \text{OPD}^2 \,/N}$$

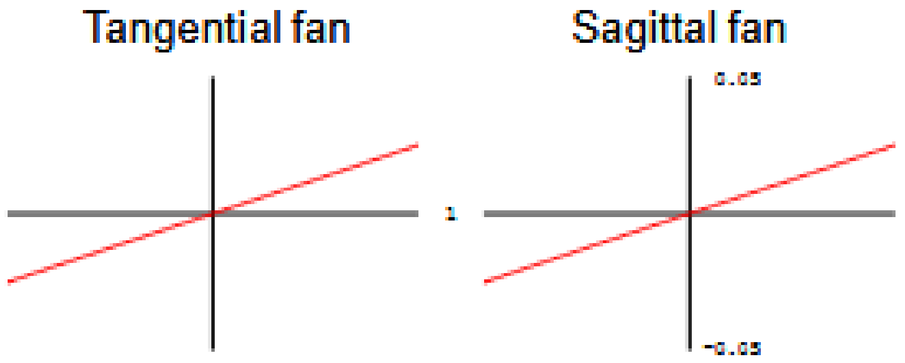


RAY ABERRATION CURVES

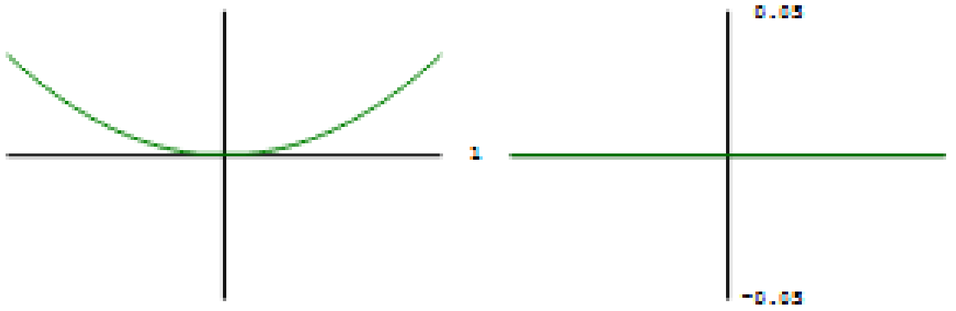




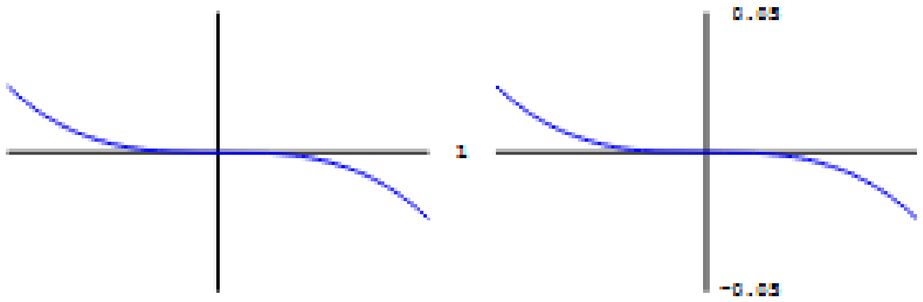
ABBERATION TYPES AND RAY ABERRATION CURVES



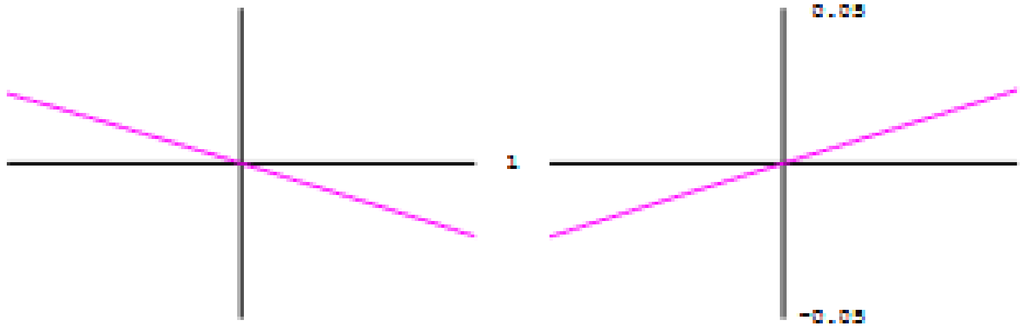
Defocus



Coma



Third-order spherical



Astigmatism

Imaging Qualitity Metrics

How to compare different imaging optical systems

SUMMARY IMAGE QUALITY METRICS

Rays or Wavefronts

- Ray aberration curves
- Spot diagrams
- Seidel aberrations
- Encircled (or ensquared) energy
- RMS Optical Path Difference (or wavefront error)
- Modulation transfer function (MTF)

Diffraction-based

- Point spread function (PSF)
- Encircled (or ensquared) energy
- MTF
- Strehl Ratio

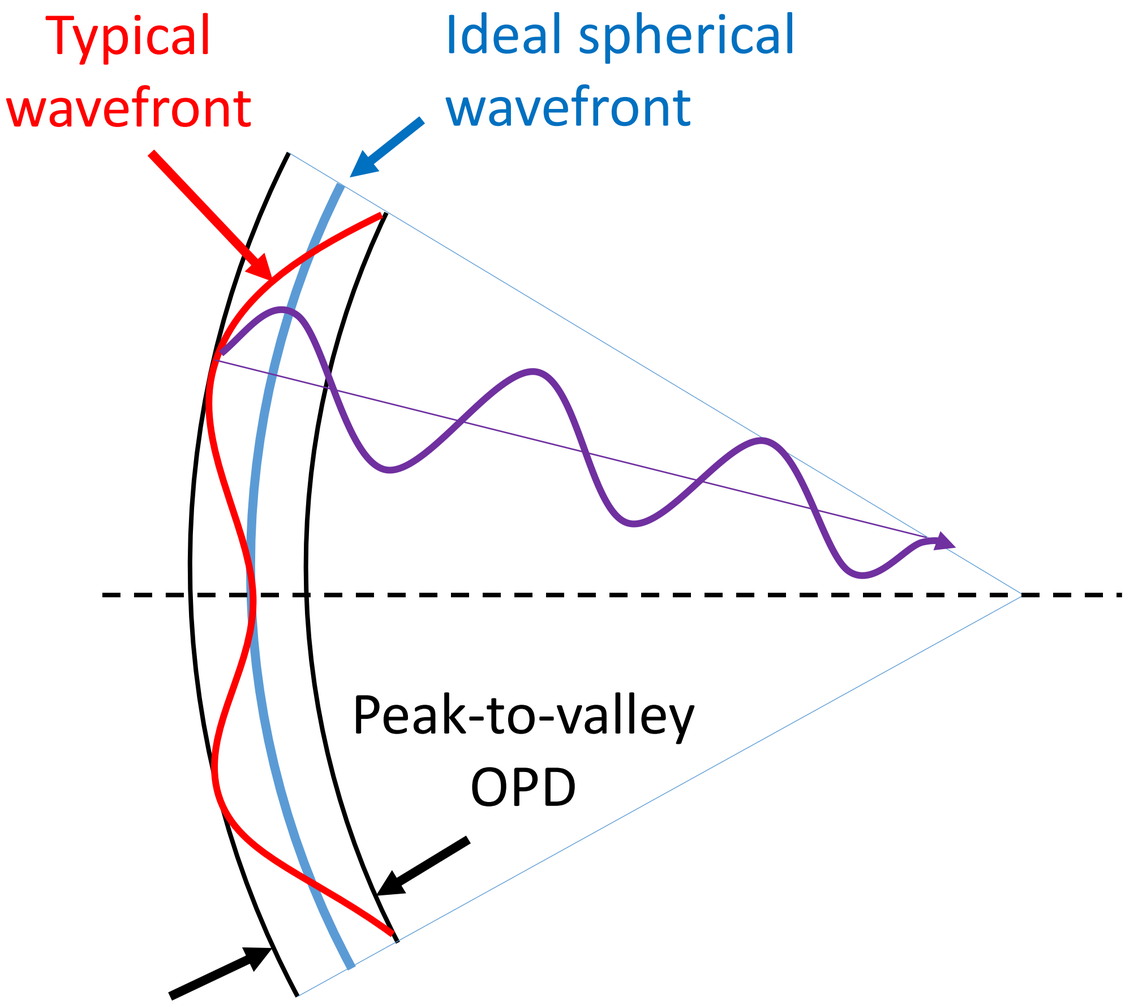
PEAK-TO-VALLEY AND RMS OPTICAL PATH DIFFERENCE

- Optical Path Difference (OPD) relates to phase difference ϕ via:

$$\text{OPD} = \frac{w}{\lambda} = \frac{2\pi\phi}{\lambda}$$

- Often wavelength units (λ)
- Typical OPD $w < \lambda$
- Peak-to-valley OPD: $\max(w)$
- RMS OPD in wavelength:

$$\sigma_w = \frac{2\pi}{\lambda}\sqrt{\langle\phi^2\rangle - \langle\phi\rangle^2}$$



PEAK-TO-VALLEY AND RMS OPTICAL PATH DIFFERENCE

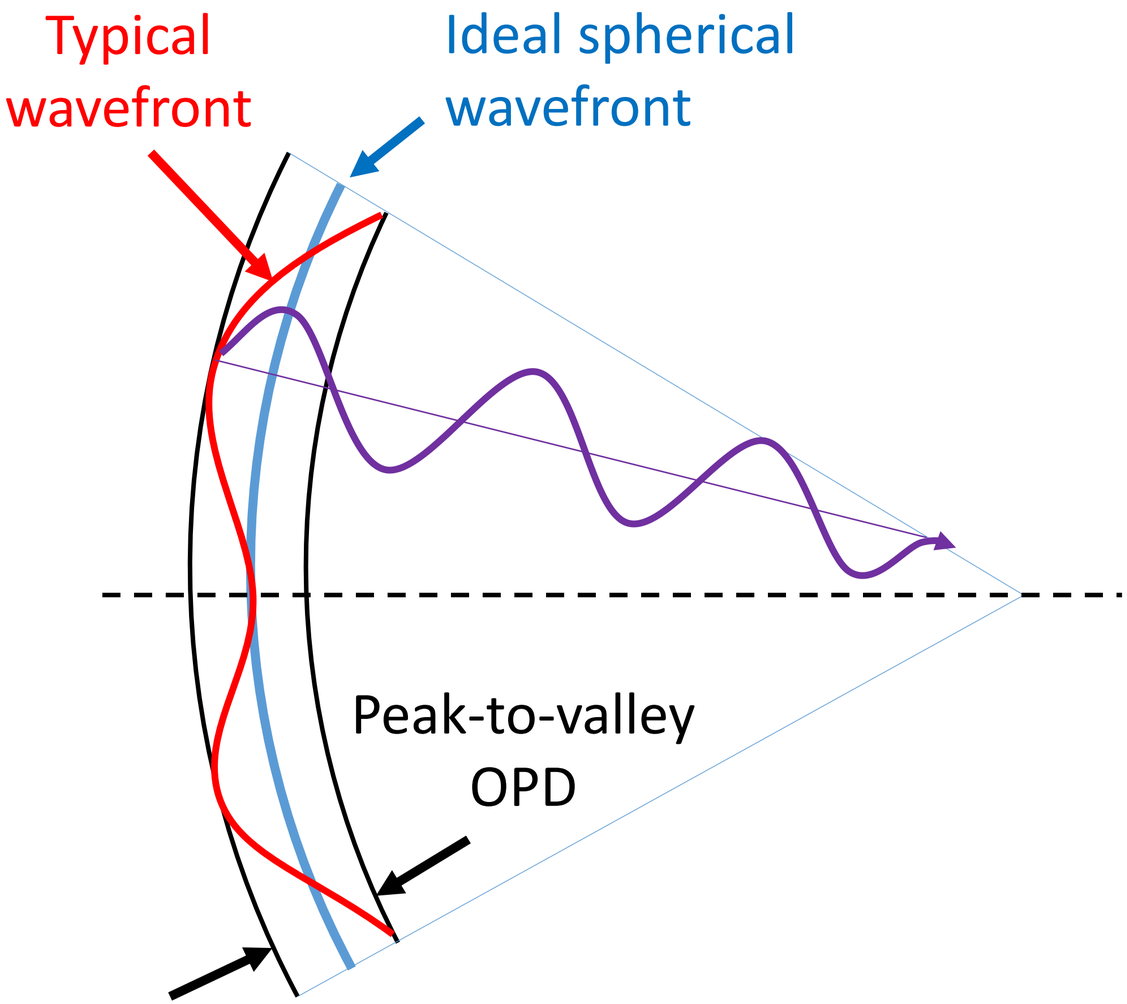
- RMS OPD < peak-to-valley

$$\sigma_w < \max(w)$$

- Combined metric:

$$\frac{\text{Peak-to-valley}}{\text{RMS}} = \frac{\max(w)}{\sigma_w}$$

Gives a sense of how well RMS OPD σ_w represents the error.

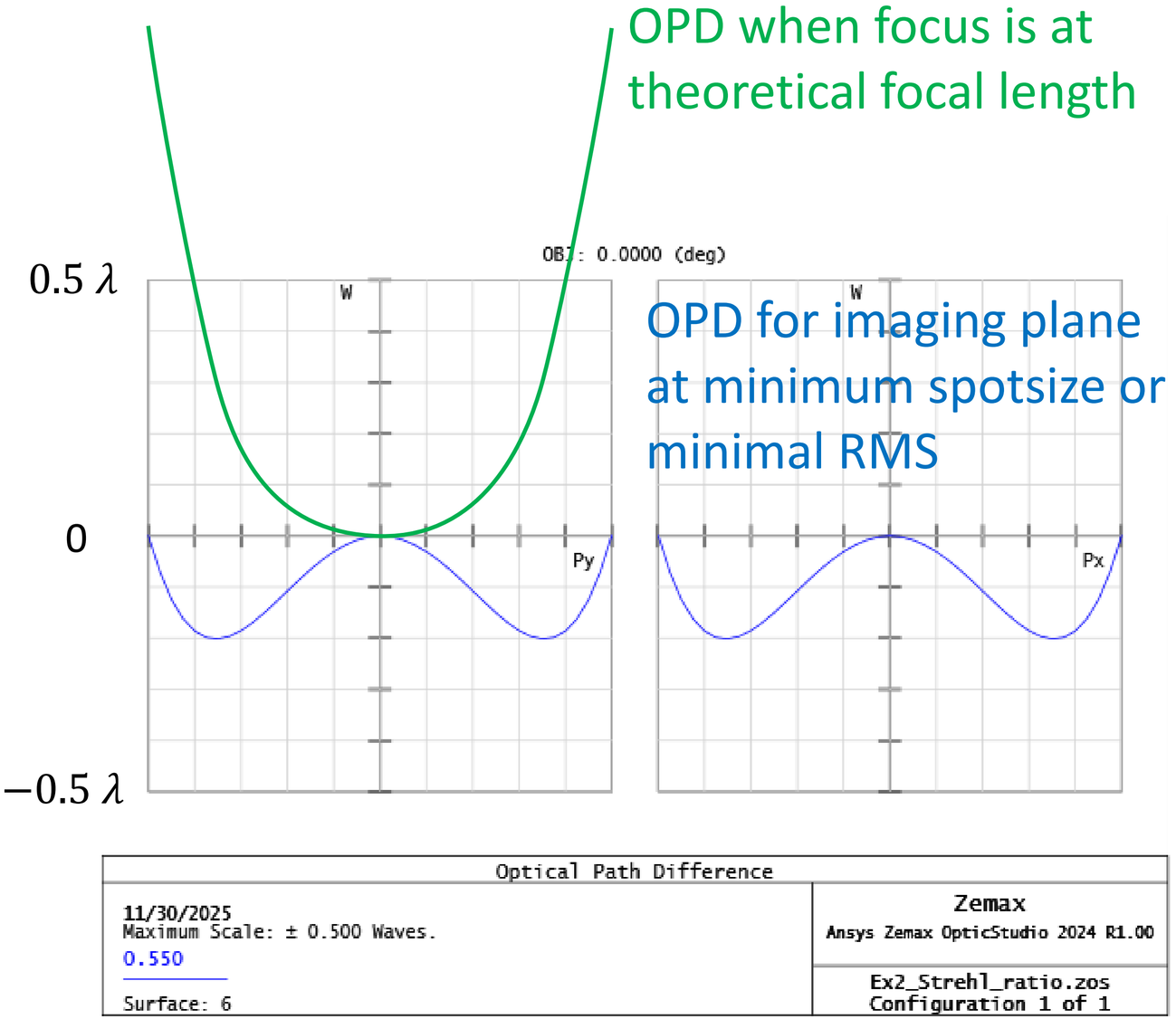


OPTICAL PATH DIFFERENCE AND IMAGE PLANE

- The optical path difference is sensitive to the position of the imaging plane
- Optimize the image plane distance first

Spherical aberration

of single biconvex lens

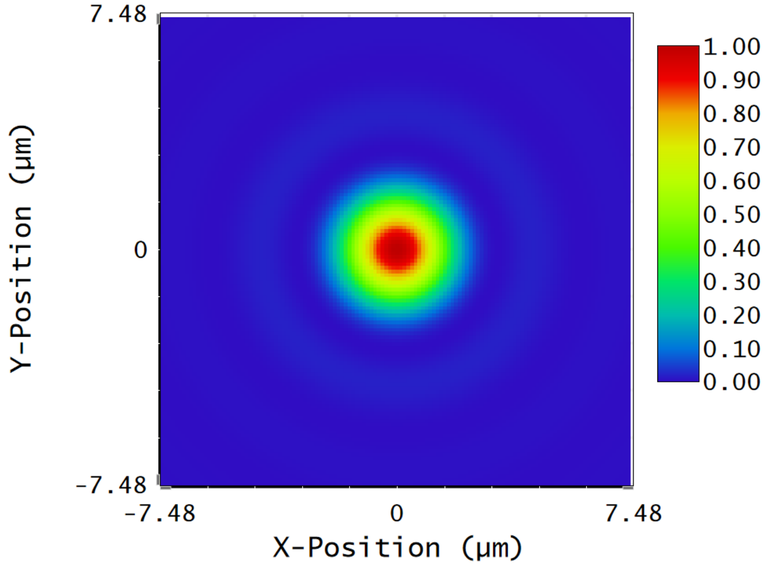


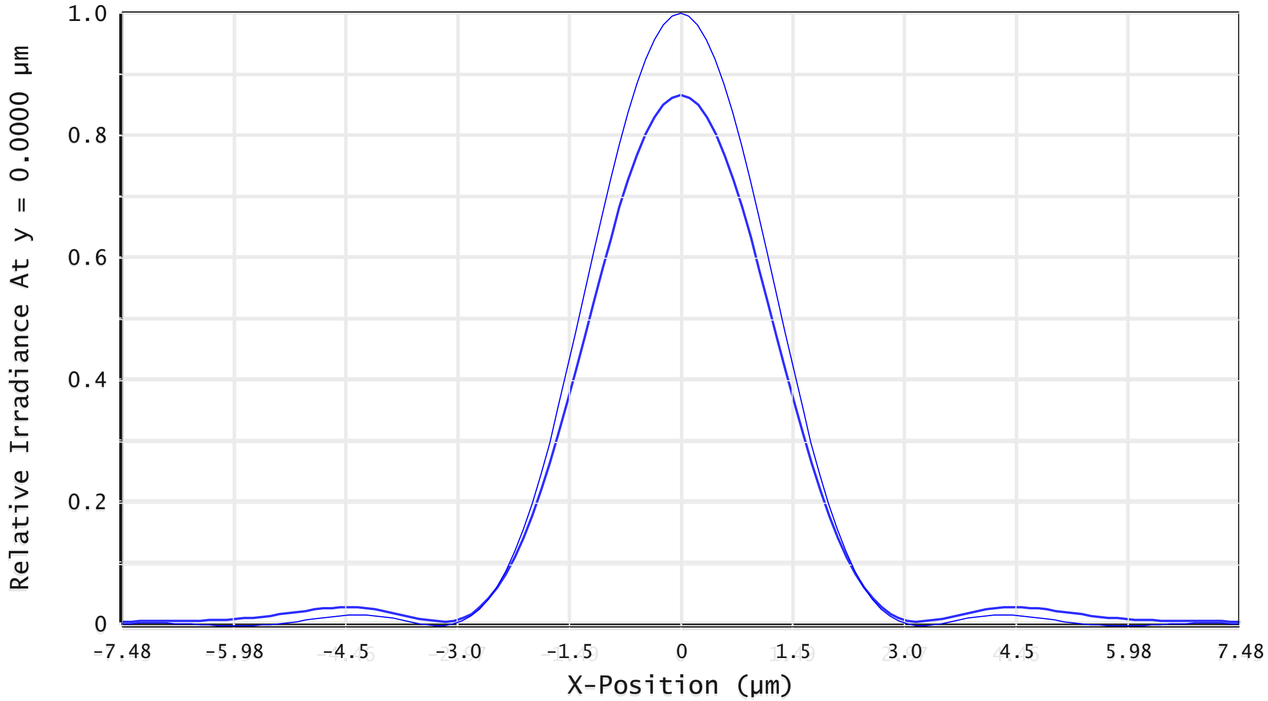
STREHL RATIO

- 1895: Dr. Karl Strehl
- Strehl ratio = peak diffraction intensity ratio:

$$S = \frac{I_{\text{peak}}}{I_{0,\text{peak}}} = \frac{I \text{ aberrated}}{I_0 \text{ perfect}}$$

- Relates PSF with aberrations
- Disadvantage:
  - Difficult to calculate





STREHL RATIO

- Strehl ratio:

$$S = \frac{I_{\text{peak}}}{I_{0,\text{peak}}} \approx e^{-\sigma_\phi^2}$$

Accurate for RMS wavefront errors

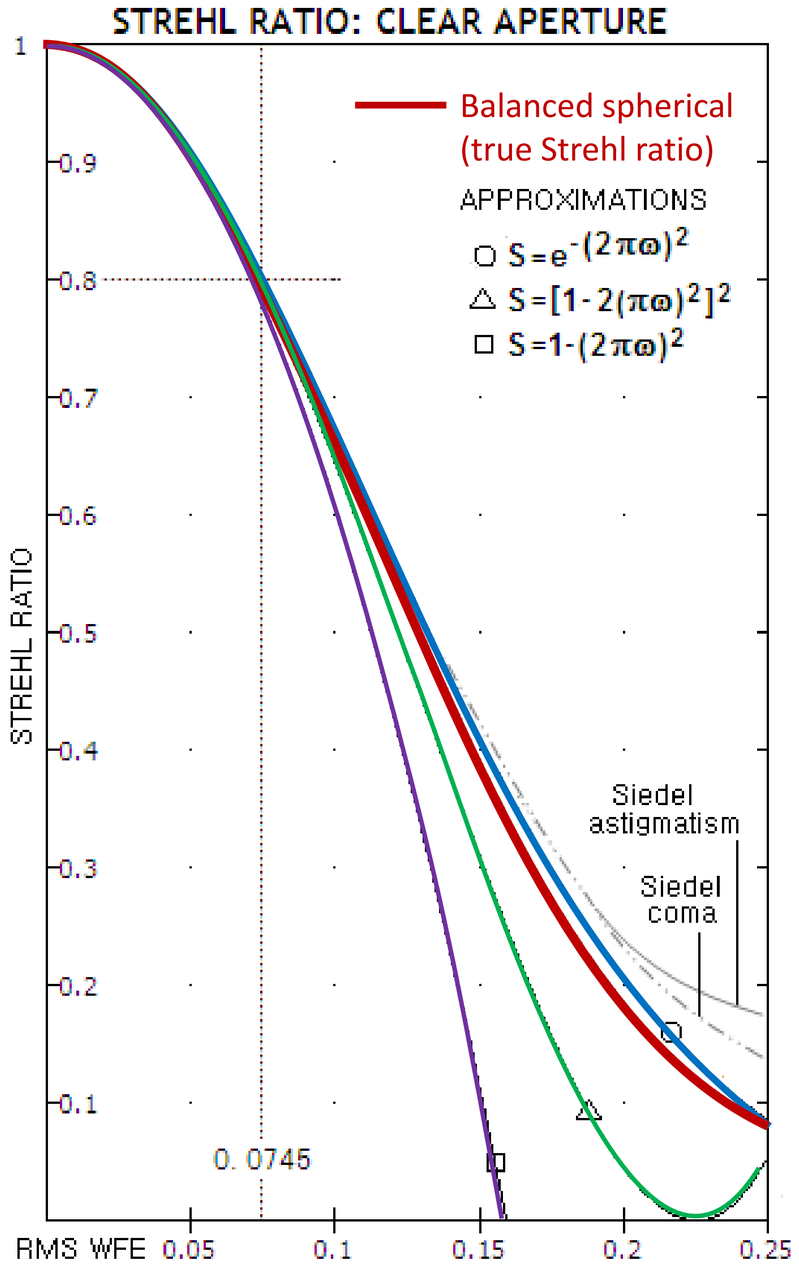
$$\Delta w \leq \lambda/4$$

Where $\sigma_\phi^2 = \left(\frac{2\pi}{\lambda}\right)^2 \Delta w^2$

Δw = RMS OPD in radians/wave

$$\approx \left[1 - \tfrac{1}{2}(2\pi w)^2\right]^2 \qquad \Delta w \leq \lambda/10$$

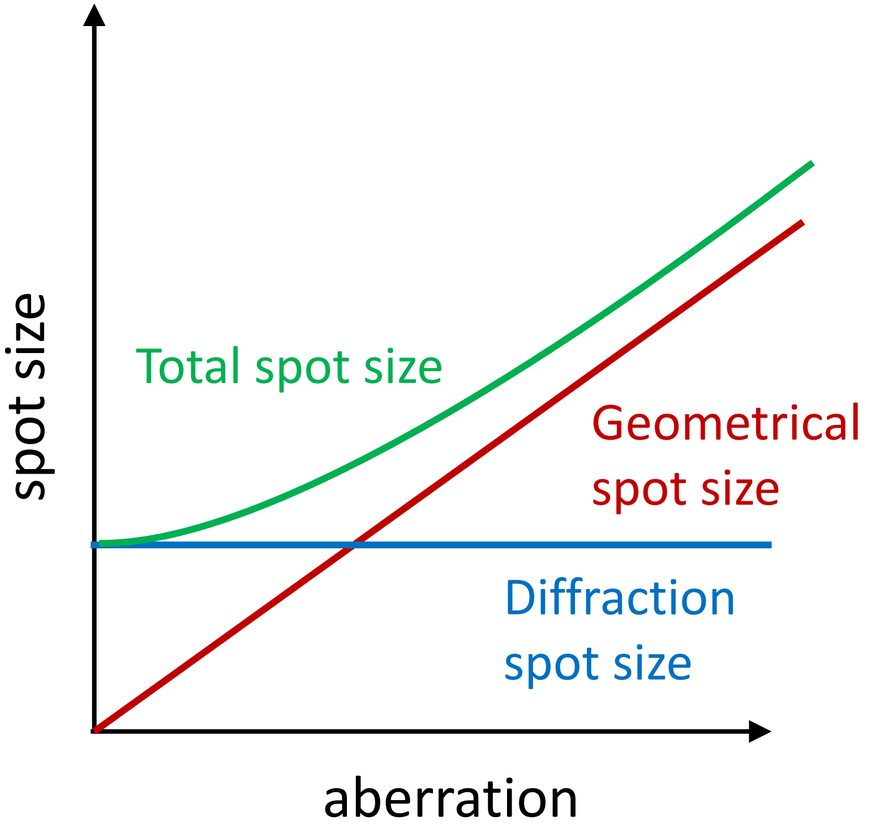
$$\approx 1 - (2\pi w)^2 \qquad \Delta w \leq \lambda/15$$



Adapted from: https://www.telescope-optics.net/Strehl.htm

STREHL RATIO & DIFFRACTION LIMITED SYSTEMS

- Strehl ratio: How well does the system approximate the “perfect” diffraction limited system for a given aperture
- S becomes better when reducing aperture of the system
- Reducing the aperture does reduce the resolution (increase the PSF)
- A “diffraction limited system” has limited geometrical aberration contribution to total spot size



Total spot size = Diffraction spot size + geometrical spot size

Chromatic Aberration

CHROMATIC ABERRATION

- The refractive index can depend on the wavelength

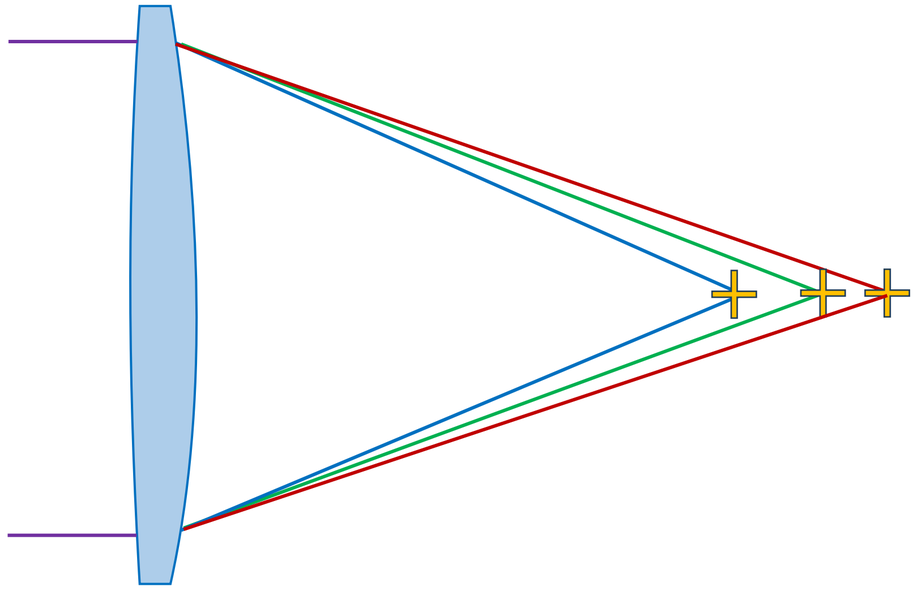
  ➡ **Dispersion** of the material

- **Abbe ν-number**:

$$\nu = \frac{(n_{\text{middle}} - 1)}{n_{\text{short}} - n_{\text{long}}}$$

- 3 wavelengths depend on material
- For glasses - visible light:

  F (486 nm), d (588 nm), C (656 nm)



Short wavelengths have most often higher refraction indices resulting in shorter focal lengths

CHROMATIC ABERRATION: APPROXIMATIONS DISPERSION

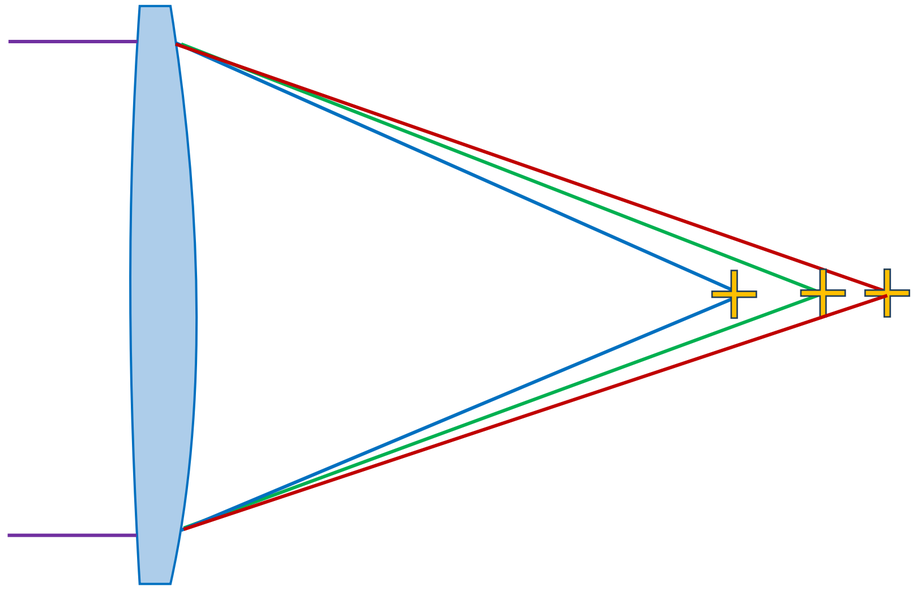
- Dispersion curve: all information

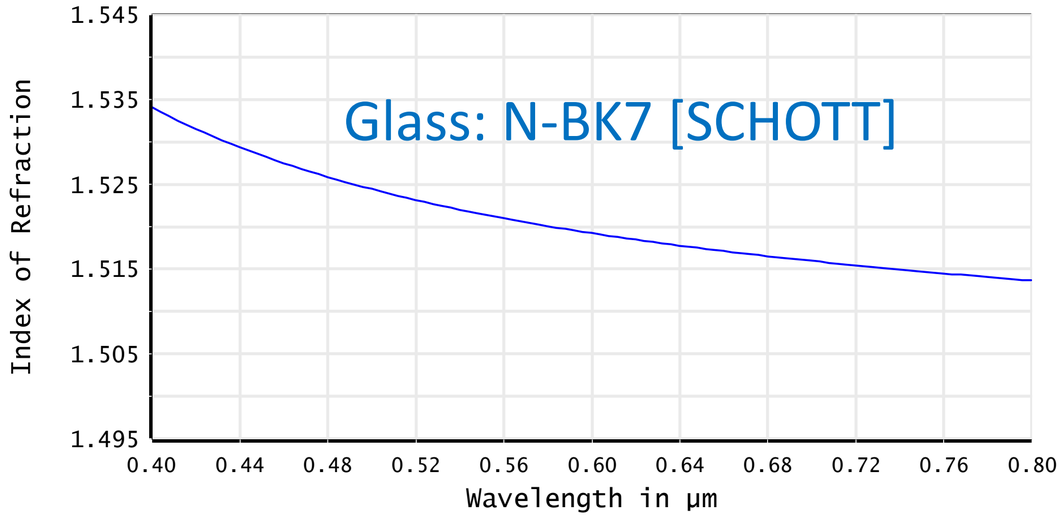
- Abbe ν-number gives average

$$\nu = \frac{(n_{\mathrm{middle}} - 1)}{n_{\mathrm{short}} - n_{\mathrm{long}}}$$

- Refraction index by linear interpolation for small deviations $\Delta\lambda$

$$n \approx n_0 + \Delta\lambda \, \frac{\mathrm{d}n(\lambda)}{\mathrm{d}\lambda}$$





Theory Seidel Aberrations

SEIDEL ABERRATIONS: 3RD ORDER ABERRATIONS

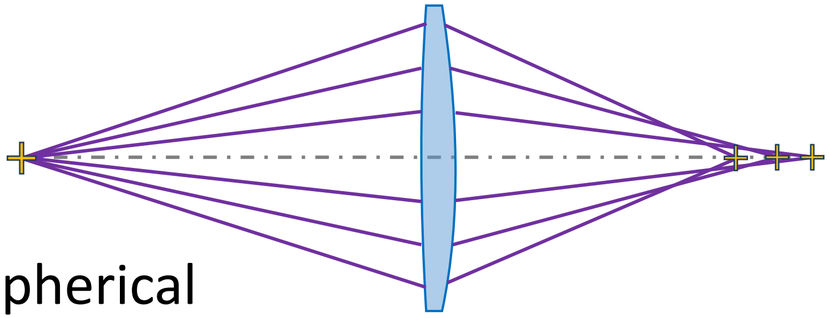
- Seidel aberrations are the 5 standard aberrations
  (1) Spherical,
  (2) Coma,
  (3) Astigmatism,
  (4) Field curvature
  (5) Distortion

- Additional: Defocus + Chromatic aberration

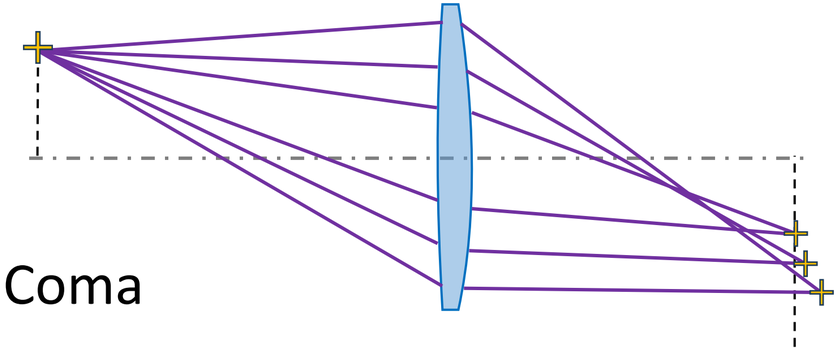
- Obtained via lowest order Taylor expansion (3rd order in pupil & object coordinates) of wavefront aberrations

- **Advantages**: Per surface analysis, additive property, intuitive
- **Disadvantage**: Higher order aberrations can matter!

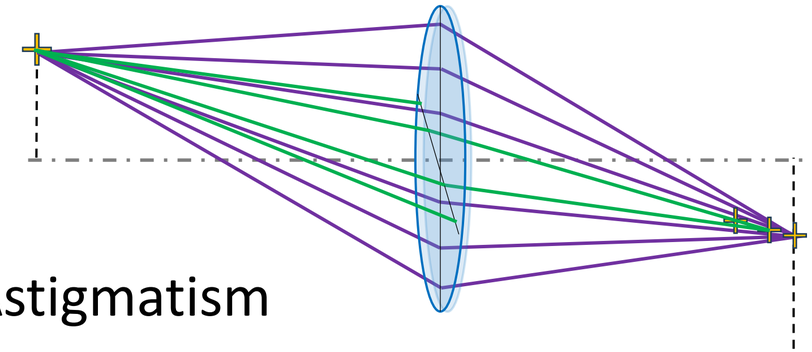
SEIDEL ABERRATIONS



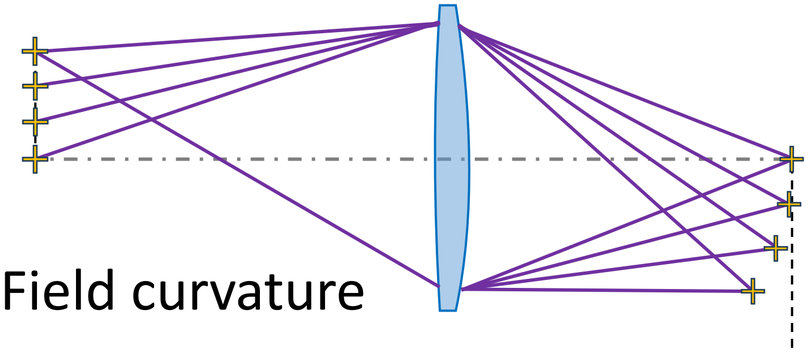
Spherical



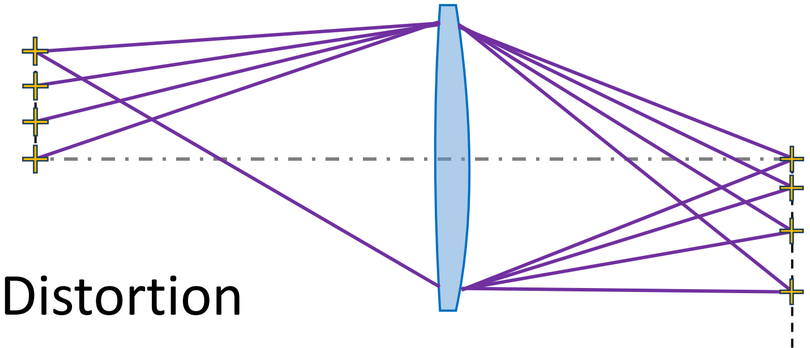
Coma



Astigmatism



Field curvature



Distortion

Rules of Thumb & Criteria

Deciding on what is “good” image quality

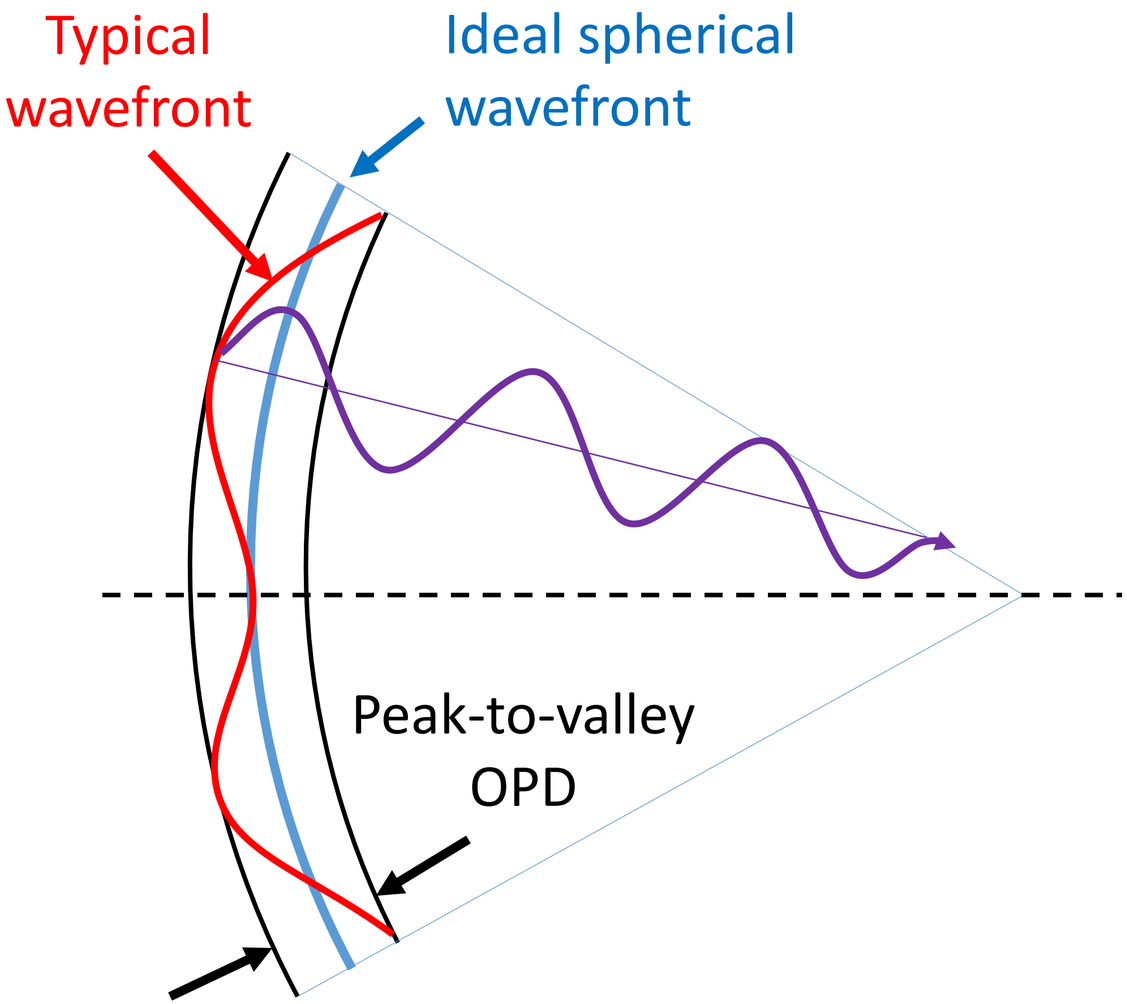
RAYLEIGH CRITERION

- Lens has “good” image quality when:

$$\text{Peak-to-valley OPD} < \frac{\lambda}{4}$$

- Same as: $\lambda/4$ optical path difference at focus

- Disadvantages:
  - ignores (non)smoothness
  - Ignores average error



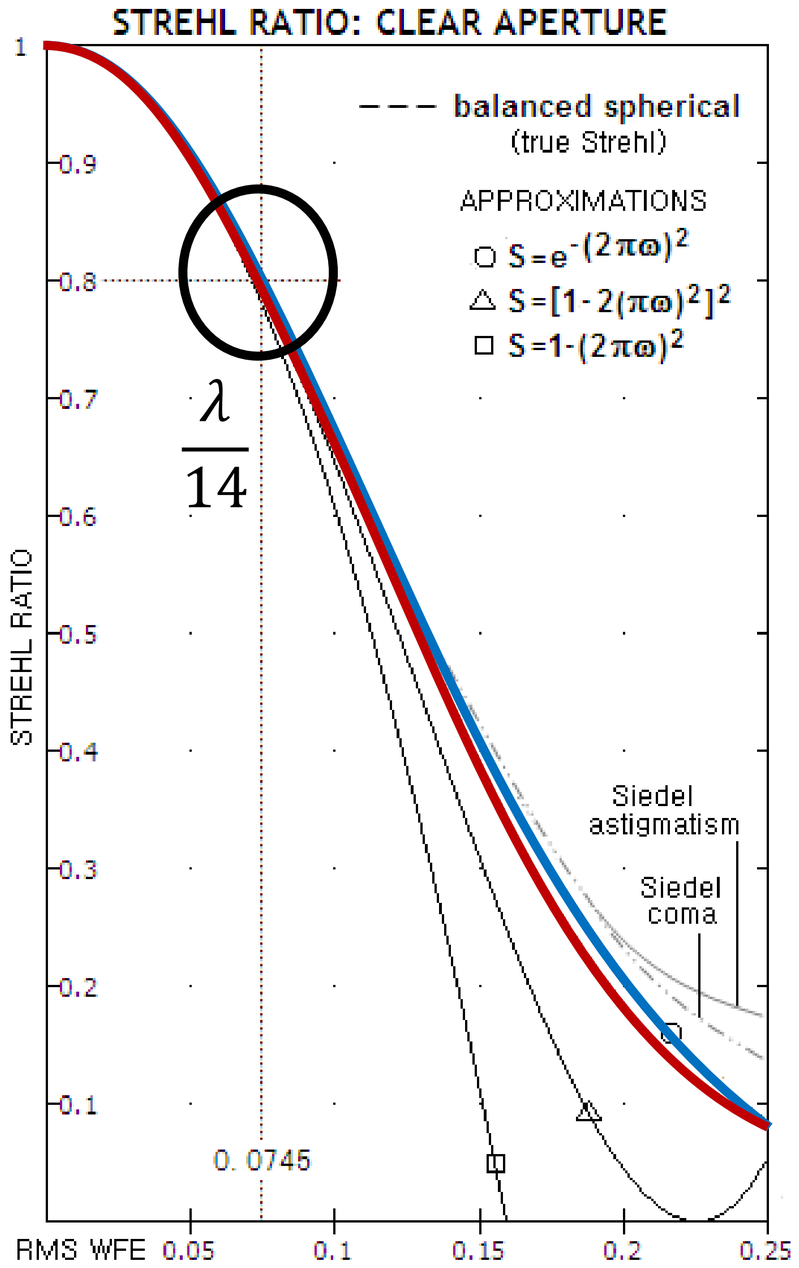
MARECHAL CRITERION

- Minimum Strehl ratio = 0.82

$$S \geq 0.82 \approx \mathrm{e}^{-\sigma_\phi^2} = e^{-\left(\frac{2\pi}{\lambda}\right)^2 \Delta w^2}$$

- Approx. maximum RMS OPD is $\frac{\lambda}{14}$

- More accurate than Rayleigh criterion
- Disadvantage: ignores smoothness of the OPD



Adapted from: https://www.telescope-optics.net/Strehl.htm

Geometric Optics Practical session